Diseases Of The Peripheral Nervous System

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Learning objectives

- Different mechanisms of damage to Peripheral Nerves.
- Definitions of different types of neuropathy.
- Clinical presentations of different and important neuropathy.
- Clinical presentation and management of AIDP.

- Numerous inherited and acquired pathological processes may affect the nerve roots (radiculopathy), the nerve plexuses (plexopathy) and/or the individual nerves (neuropathy).
- Cranial nerves 3-12 share the same tissue characteristics as peripheral nerves elsewhere and are subject to the same range of diseases.
- Nerve fibers of different types (motor, sensory or autonomic) and of different sizes may be variably involved.
- Disorders may be primarily directed at the axon, the myelin sheath (Schwann cells) or both.

Mechanisms Of Damage To Peripheral Nerves

- Peripheral nerves consist of two principal cellular structures - the axon and its anterior horn cell, and the myelin sheath, produced by Schwann cells between each node of Ranvier.
- Six principal mechanisms, some coexisting, cause nerve malfunction:
- **Demyelination:** When the Schwann cell is damaged, the myelin sheath is disrupted. This causes marked slowing of conduction, seen for example in Guillain-Barré syndrome, post-diphtheritic neuropathy and many hereditary sensorimotor neuropathies

- **Axonal degeneration**: The primary damage is in the axon, which dies back from the periphery. Conduction velocity tends to remain normal. Axonal degeneration occurs typically in toxic neuropathies.
- **Walleriandegeneration:** This describes the changes following nerve section. Both axon and distal myelin sheath degenerate over several weeks.
- **Compression:** Focal demyelination at the point of compression. The myelin sheath is disrupted. This occurs typically in entrapment neuropathies e.g. carpal tunnel syndrome.

- **Infarction:** Microinfarction of *vasa nervorum* occurs in arteritis, e.g. polyarteritis nodosa, and in diabetes mellitus. Wallerian degeneration occurs distal to the ischaemic zone.
- **Infiltration:** Peripheral nerves are infiltrated by inflammatory cells in leprosy, by malignant cells or granulomas in sarcoidosis.

Nerve Regeneration

- Regeneration occurs either by remyelination, when recovering Schwann cells spin new myelin sheaths around the axon, or by axonal growth down the nerve sheath and sprouting from the axonal stump.

- Axonal growth takes place at a rate of up to 1 mm daily.

Definitions

- **Neuropathy** means a pathological process affecting a peripheral nerve or nerves.
- **Mononeuritis Simplex** means a process affecting a single nerve.
- **Mononeuritis multiplex** means a process affecting several or multiple nerves usually sequentially.
- **Polyneuropathy** describes diffuse, symmetrical disease, usually beginning peripherally. The course is either acute, chronic, static, progressive, relapsing or towards recovery. Polyneuropathies are motor, sensory, sensorimotor (i.e. mixed) and autonomic. They are classified broadly into demyelinating and axonal types, depending upon which principal pathological process predominates.

- **Radiculopathy** means disease affecting nerve roots.
- **Plexopathy** means a disease affecting the nerve plexuses
- Diagnosis is made by clinical pattern, nerve conduction studies, nerve biopsy (usually sural or radial nerve) and identification of systemic or genetic disease.

Focal Neuropathy (Mononeuropathy)

Entrapment neuropathy ➔ is the cause.

Pressure ➔ damages the myelin sheath, and neurophysiology will show slowing of conduction over the relevant site.

Sustained or severe pressure ➔ axons, demonstrable as loss of the sensory action potential distal to the site of compression.

➔ single nerve lesion ➔ later ➔ multiple nerve lesions (clinically or neurophysiologically), i.e. a multifocal neuropathy (mononeuritis multiplex).

Certain conditions increase the propensity to develop entrapment neuropathies.

- Acromegaly
- Obesity
- Hypothyroidism
- Pregnancy
- Any pre-existing mild generalised axonal neuropathy (e.g. diabetes)
- Bone damage near the nerve

Symptoms And Signs In Common Entrapment Neuropathies

Nerve	Symptoms	Muscle weakness/muscle-wasting	Area of sensory loss
Median (at wrist) (carpal tunnel syndrome)	Pain and paraesthesia on palmar aspect of hands and fingers, waking the patient from sleep. Pain may extend to arm and shoulder	Abductor pollicis brevis	Lateral palm and thumb, index, middle and lateral half 4th finger
Ulnar (at elbow)	Paraesthesia on medial border of hand, wasting and weakness of hand muscles	Small muscles of the hand	Medial palm and little finger, and medial half 4th finger

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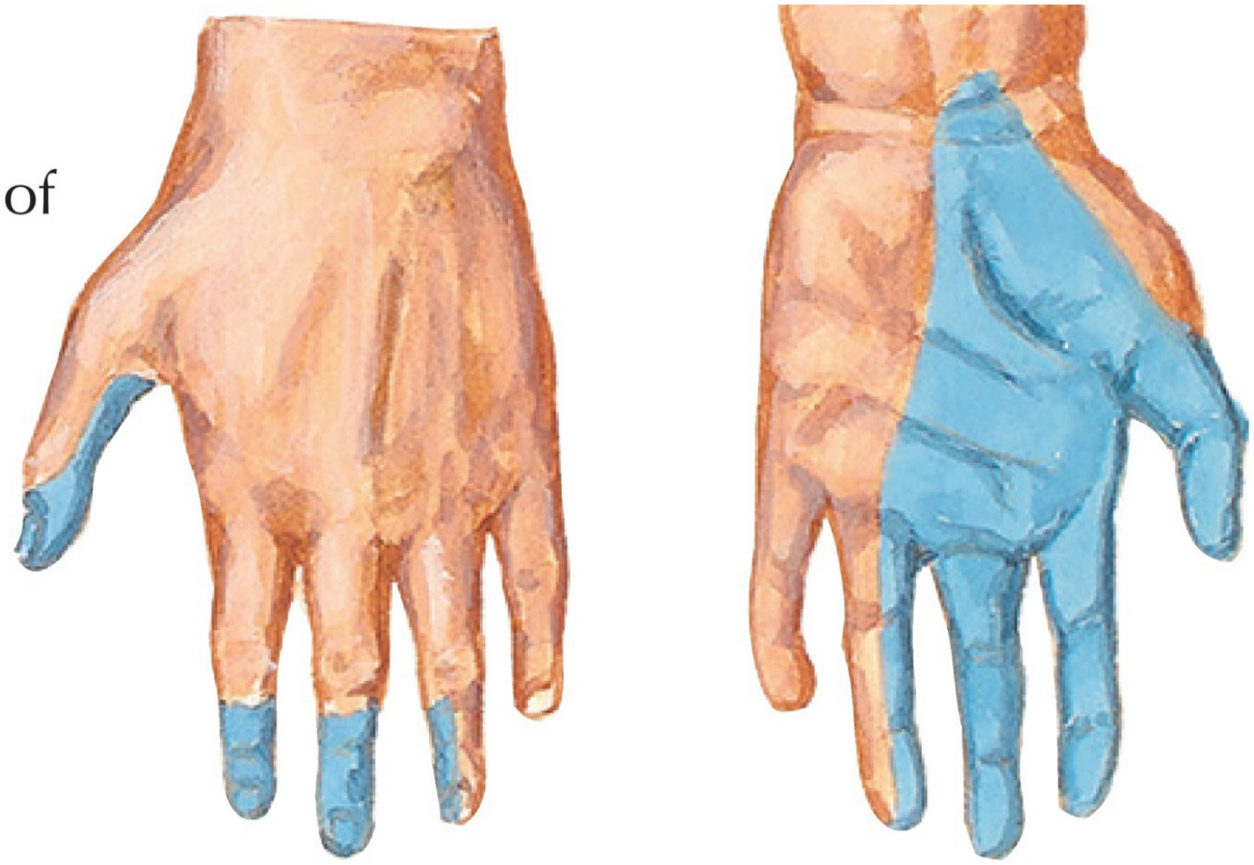
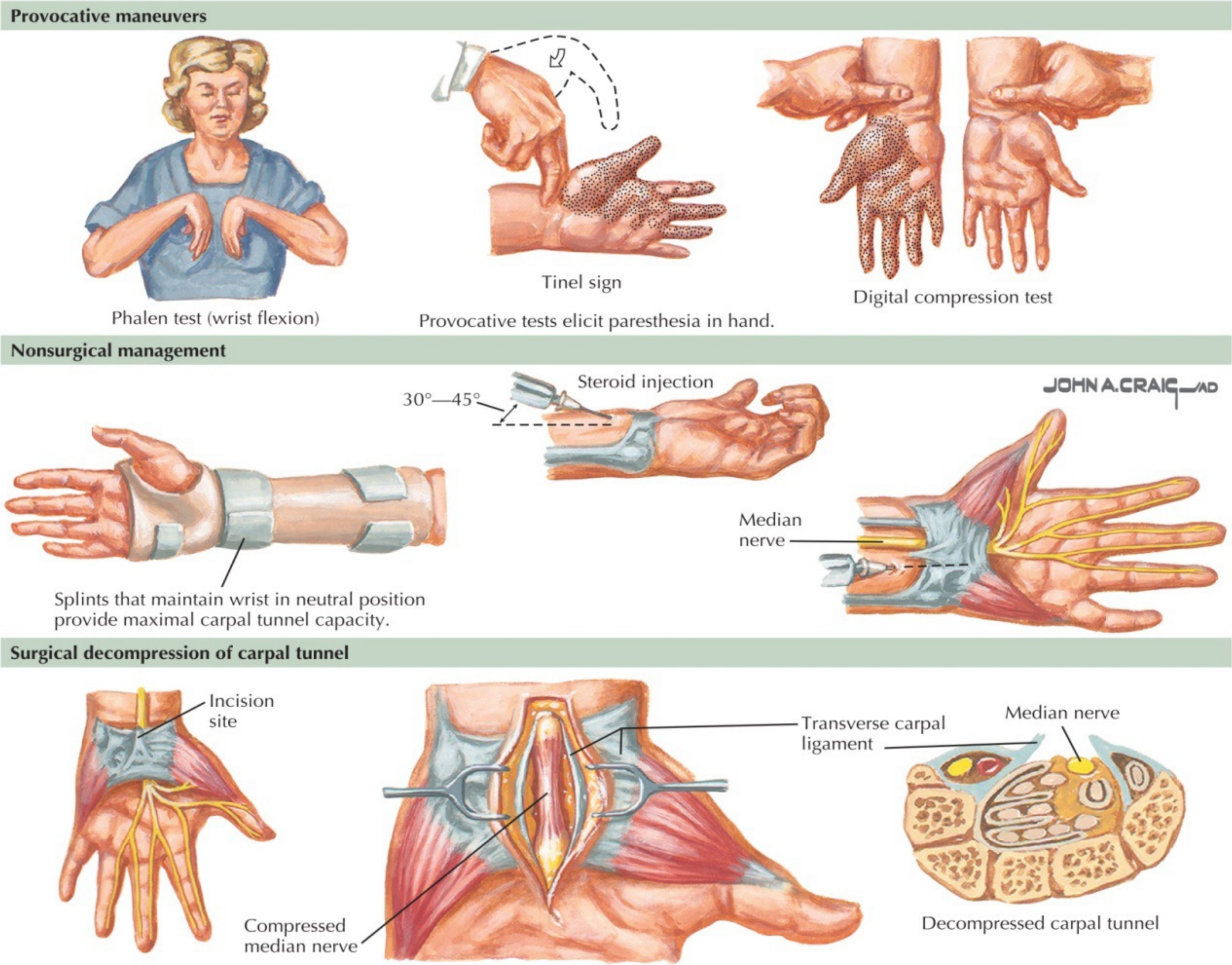
Median Nerve

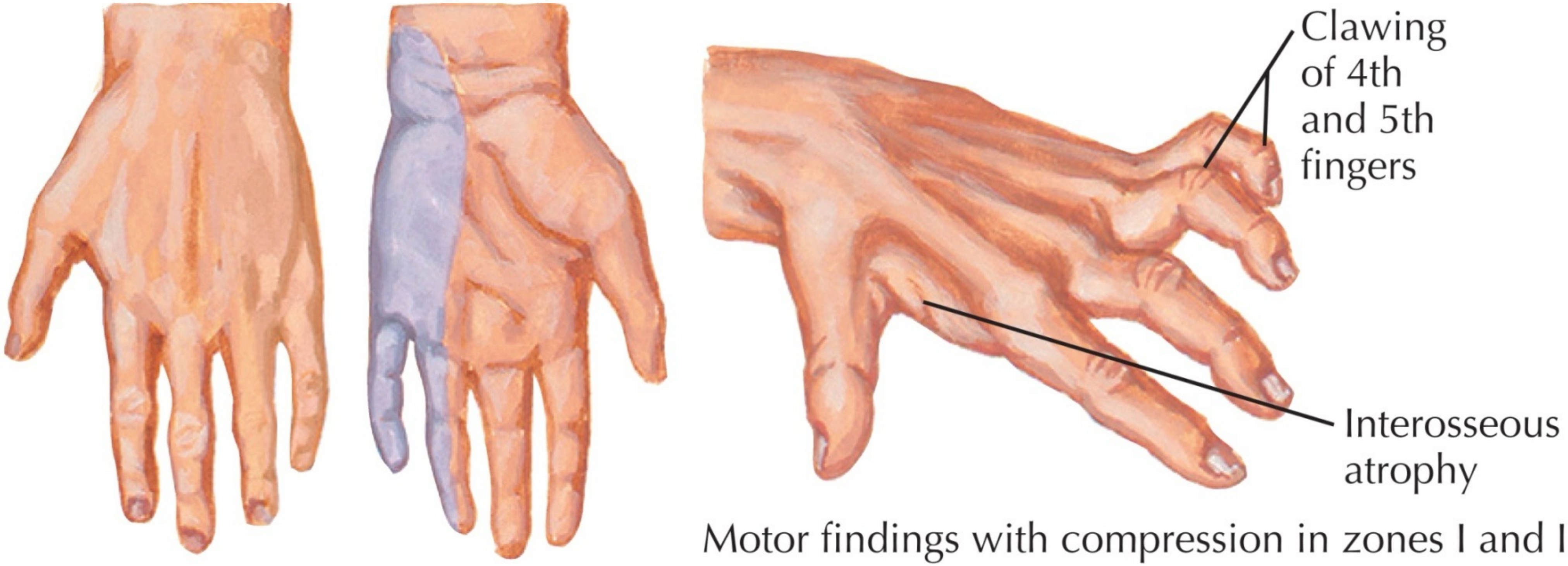
Sensory distribution of median nerve



Carpal Tunnel Syndrome

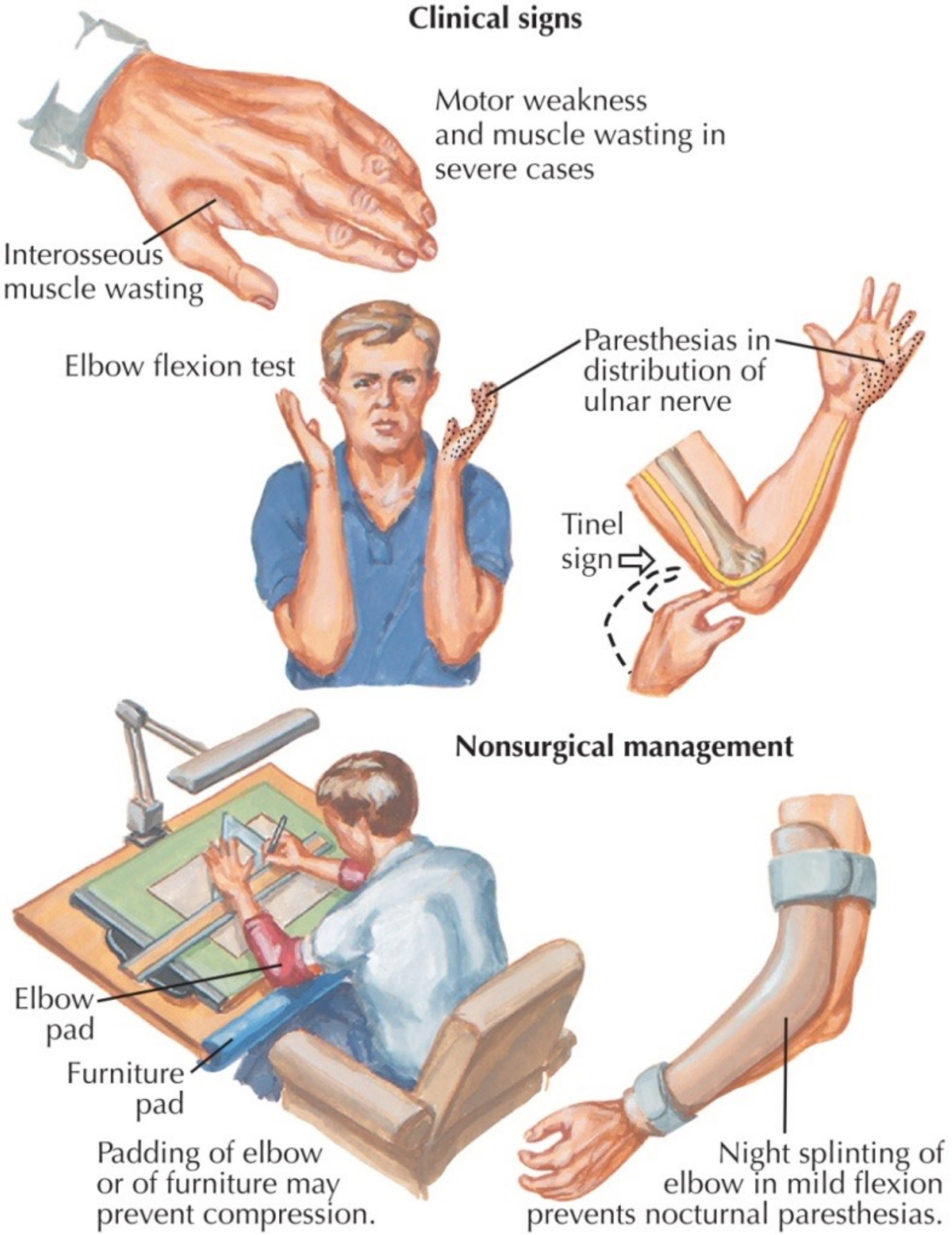


Ulnar Tunnel Syndrome



Motor findings with compression in zones I and II

Cubital Tunnel Syndrome



Nerve	Symptoms	Muscle weakness/muscle-wasting	Area of sensory loss
Radial	Weakness of extension of wrist and fingers, often precipitated by sleeping in abnormal posture, e.g. arm over back of chair	Wrist and finger extensors	Dorsum of thumb
Peroneal	Foot drop, trauma to head of fibula	Dorsiflexion and eversion of foot	dorsum of foot

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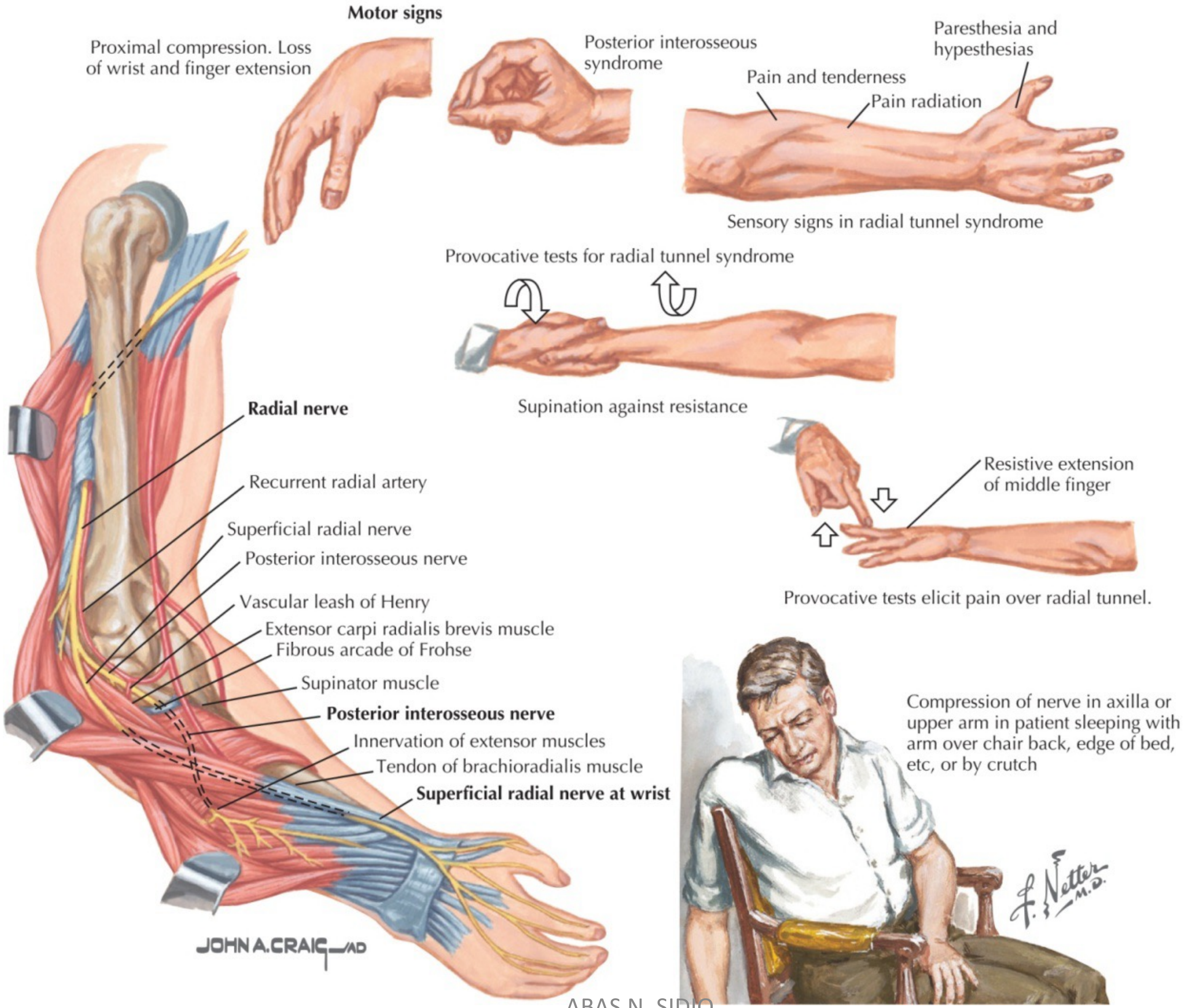
Radial Tunnel Syndrome



Idiopathic isolated facial nerve palsy **Bell's palsy**

all ages and both sexes.

The lesion is within the facial canal and may be due to reactivation of latent **herpes simplex virus 1** infection.

develop subacutely over a few hours.

pain around the ear preceding the unilateral facial weakness.

Patients often describe the face as 'numb', but there is no objective sensory loss (except possibly to taste).

- Hyperacusis ➔ nerve to stapedius is involved, and there may be diminished salivation and tear secretion.
- Examination reveals an ipsilateral lower motor neuron facial nerve palsy.
- Vesicles in the ear or on the palate indicate that the facial palsy is due to **herpes zoster** rather than Bell's palsy(Ramsay Hunt syndrome or Herpes zoster oticus) :
- This is herpetic inflammation of the Gasserian (geniculate) ganglion causing zoster of the outer ear or oropharynx with acute facial (VII) nerve palsy, with or without acoustic nerve involvement (tinnitus, vertigo, deafness).

Right Bell's Palsy

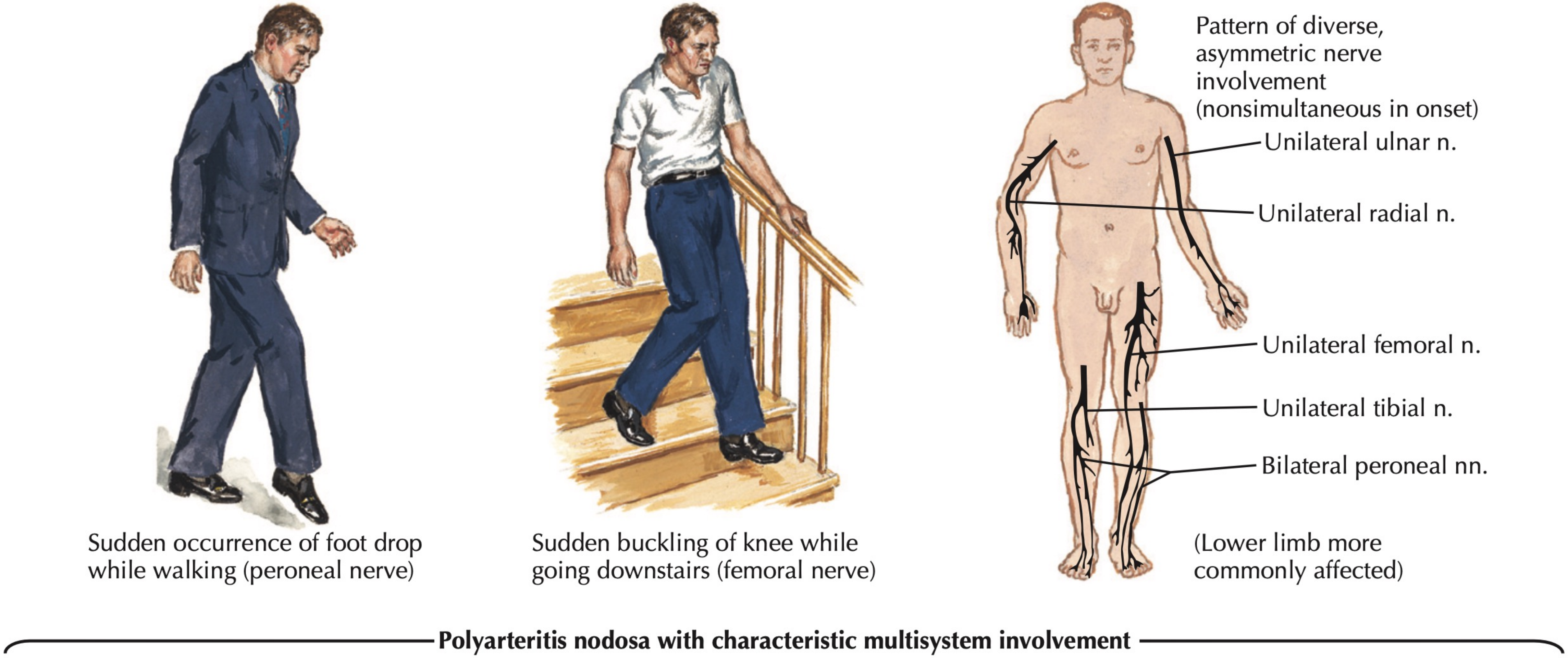
Ramsay Hunt syndrome

Treatment of Bell's Palsy

- Prednisolone 40-60 mg daily for a 1-2 weeks may speed recovery if started within 72 hours, and acyclovir has also been recommended.
- Artificial tears and ointment prevent exposure keratitis and the eye should be taped shut overnight.
- About 80% of patients recover spontaneously within 12 weeks.
- A slow or poor recovery is predicted by complete paralysis, older age and reduced facial motor action potential amplitude after the first week.

Multifocal Neuropathy (Mononeuritis Multiplex)

- When multiple nerve root, peripheral nerve or cranial nerve lesions occur serially or concurrently, the pathology is due either to involvement of the vasa nervosum or to malignant infiltration of the nerves.
- widespread multifocal neuropathy ➔ may eventually resembles a polyneuropathy.
- In this case neurophysiology may be required to identify the multifocal nature of the problem.
- Investigation should be urgent since vasculitis is a common cause, either as part of a systemic disease or isolated to the nerves.



Sudden occurrence of foot drop while walking (peroneal nerve)
Sudden buckling of knee while going downstairs (femoral nerve)
Pattern of diverse, asymmetric nerve involvement (nonsimultaneous in onset)
Unilateral ulnar n.
Unilateral radial n.
Unilateral femoral n.
Unilateral tibial n.
Bilateral peroneal nn.
(Lower limb more commonly affected)
Polyarteritis nodosa with characteristic multisystem involvement

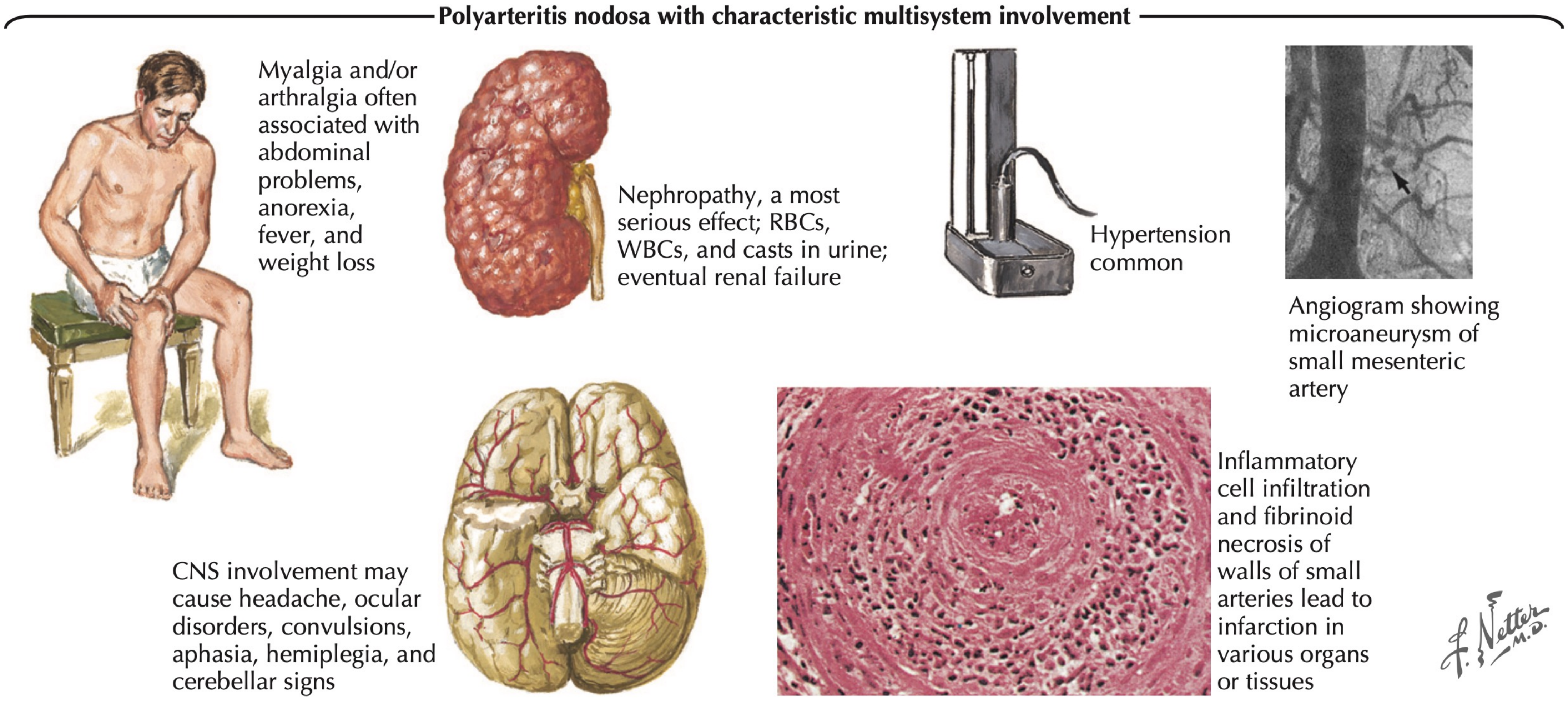


Figure 72-5 Mononeuritis Multiplex due to Systemic Vasculitis.

Polyneuropathy

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- The clinical effects of a generalized pathological process occur in the longest peripheral nerves first, affecting the distal lower limbs before the upper limbs, with sensory symptoms and signs of an ascending 'glove and stocking' distribution .
- This is particularly true with axonal neuropathies where the disorder affects the metabolic processes required for axonal transport in the peripheral nerves.
- In inflammatory demyelinating neuropathies, the pathology may be patchier and variations from this ascending pattern occur.
- Polyneuropathy could be hereditary or acquired.

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Some Causes Of Hereditary Polyneuropathies

- Hereditary motor and sensory neuropathy (Charcot-Marie-Tooth Disease).
- Hereditary neuropathy with liability to pressure palsies.
- Hereditary sensory and autonomic neuropathy.
- Fabry disease.
- Refsum disease.
- Tangier disease.
- Transthyretin (TTR) amyloidosis.
- Gelsolin amyloidosis.
- Apolipoprotein A-I amyloidosis.

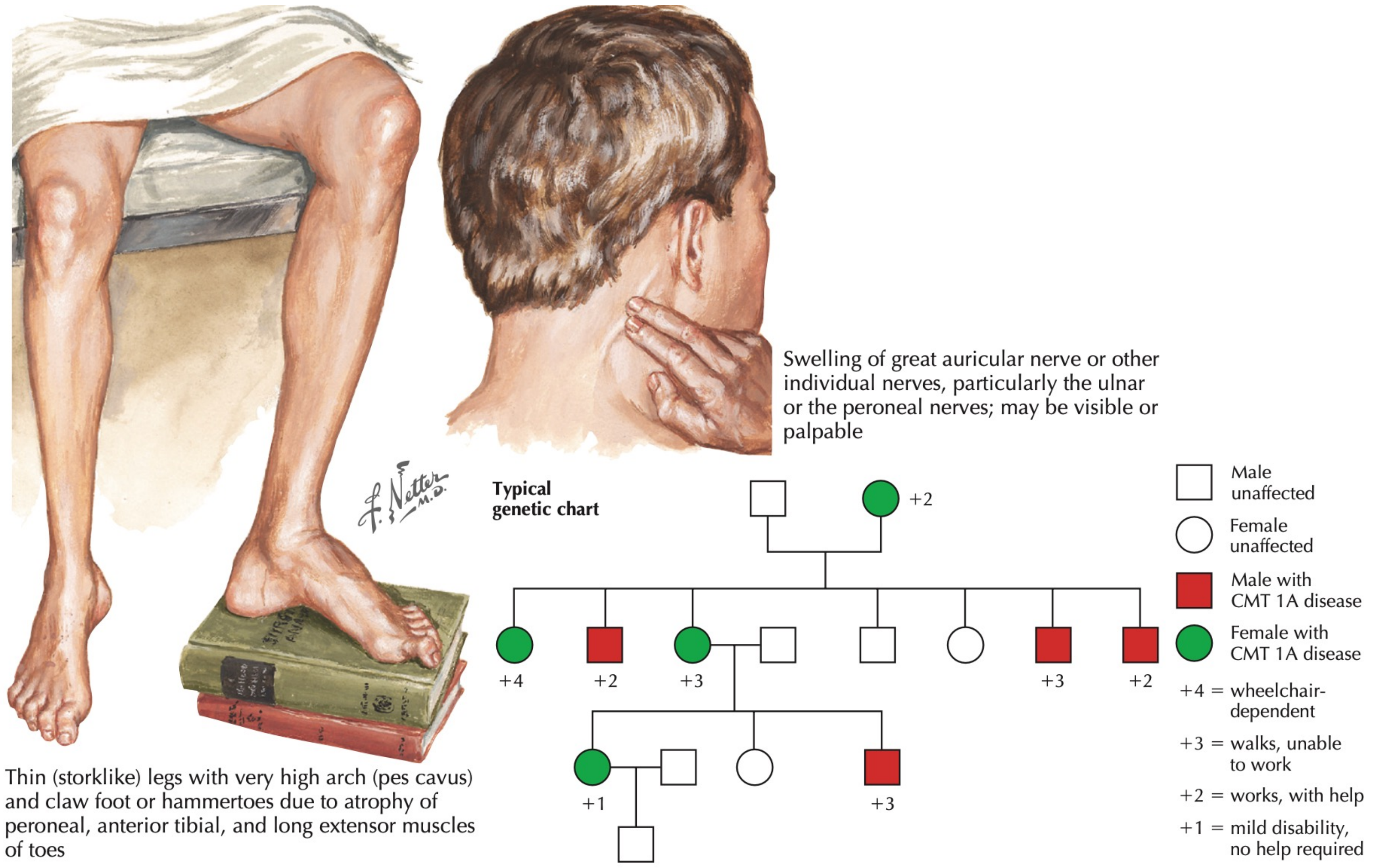


Figure 71-1 Findings in Charcot–Marie–Tooth Disease.

Acquired Polyneuropathies

- Metabolic

a) Diabetes mellitus
b) Hypothyroidism
c) Vitamin B12 deficiency
d) Copper deficiency
e) Thiamine deficiency
f) Alcohol
g) Uremia
h) Hyperlipidemia
i) Hypertriglyceridemia

Acquired Polyneuropathies

- Inflammatory

a) AIDP (Acute Inflammatory Demyelinating Polyneuropathy).

b) CIDP (Chronic Inflammatory Demyelinating Polyneuropathy)

c) Multifocal motor neuropathy.

d) Vasculitis: systemic and nonsystemic.

e) Sarcoidosis.

f) Sjögren syndrome.

- Toxic:
- Lead, Mercury, Arsenic, Thallium, Vitamin B6

Acquired Polyneuropathies

- Infectious:

a) HIV
b) Syphilis
c) Lyme disease
d) Leprosy

- Other Causes:

a) Primary amyloidosis
b) MGUS (monoclonal gammopathy of undetermined significance)
c) Paraneoplastic
d) Chemotherapy e.g. Cisplatin, Vincristine, Thalidomide etc.

Acute Inflammatory Demyelinating Polyneuropathy (AIDP OR Guillain-Barré Syndrome)

- 1-4 weeks ➔ respiratory infection or diarrhea (particularly *Campylobacter*) in 70% of patients.
- Predominantly cell-mediated inflammatory response directed at the myelin protein of spinal roots, peripheral and extra-axial cranial nerves, possibly triggered by molecular mimicry between epitopes found in the cell walls of some micro-organisms and gangliosides in the Schwann cell membranes.
- The resulting release of inflammatory cytokines blocks nerve conduction and is followed by a complement-mediated destruction of the myelin sheath and the associated axon, if it is severe.

Clinical features

- Distal paraesthesia and limb pains (often severe) precede a rapidly ascending muscle weakness, from lower to upper limbs, more marked proximally than distally.
- Facial and bulbar weakness commonly develops, and respiratory weakness requiring ventilatory support occurs in 20% of cases.
- In most patients, weakness progresses for 1-3 weeks, but rapid deterioration to respiratory failure can develop within hours.
- On examination there is diffuse weakness with widespread loss of reflexes (AREFLEXIA).

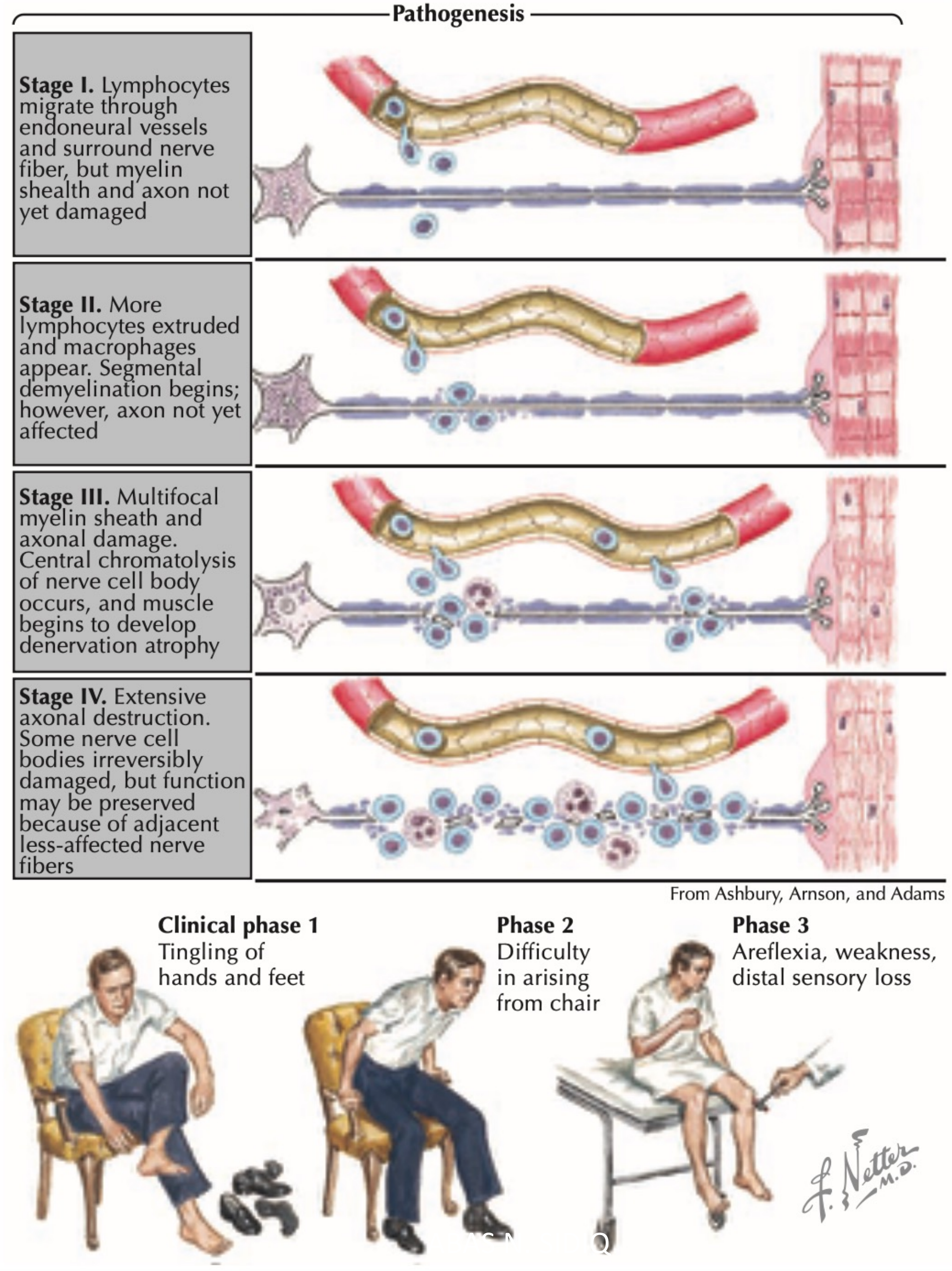
SUBTYPES OF GUILLAIN-BARRÉ SYNDROME (GBS)

SUBTYPE	FEATURES	ELECTRODIAGNOSIS
Acute inflammatory demyelinating polyneuropathy (AIDP)	Adults affected more than children; 90% of cases in Western world; recovery rapid; anti-GM1 antibodies (<50%)	Demyelinating
Acute motor axonal neuropathy (AMAN)	Children and young adults; prevalent in China and Mexico; may be seasonal; recovery rapid; anti-GD1a antibodies	Axonal
Acute motor sensory axonal neuropathy (AMSAN)	Mostly adults; uncommon; recovery slow, often incomplete; closely related to AMAN	Axonal
Miller Fisher syndrome (MFS)	Adults and children; uncommon; ophthalmoplegia, ataxia, and areflexia; anti-GQ1b antibodies (90%)	Demyelinating

Investigations

- CSF protein → elevated at some stage → may be normal in the first 10 days.
- usually no rise in CSF cell number
- Electrophysiological studies → normal early → but show typical changes after a week or so, with conduction block and multifocal motor slowing
- Investigation to identify an underlying cause, such as cytomegalovirus, mycoplasma or Campylobacter, requires a chest X-ray, stool culture and appropriate immunological blood tests.

AIDP (Guillain-Barré syndrome)



Management

- During the phase of deterioration, regular monitoring of respiratory function (vital capacity and arterial blood gases) is required, as respiratory failure may develop with little warning and require ventilatory support.
- Ventilation may be needed if the vital capacity falls below 1 litre, but intubation is more often required because of bulbar incompetence leading to aspiration.
- General management to protect the airway and prevent pressure sores and venous thrombosis is essential.

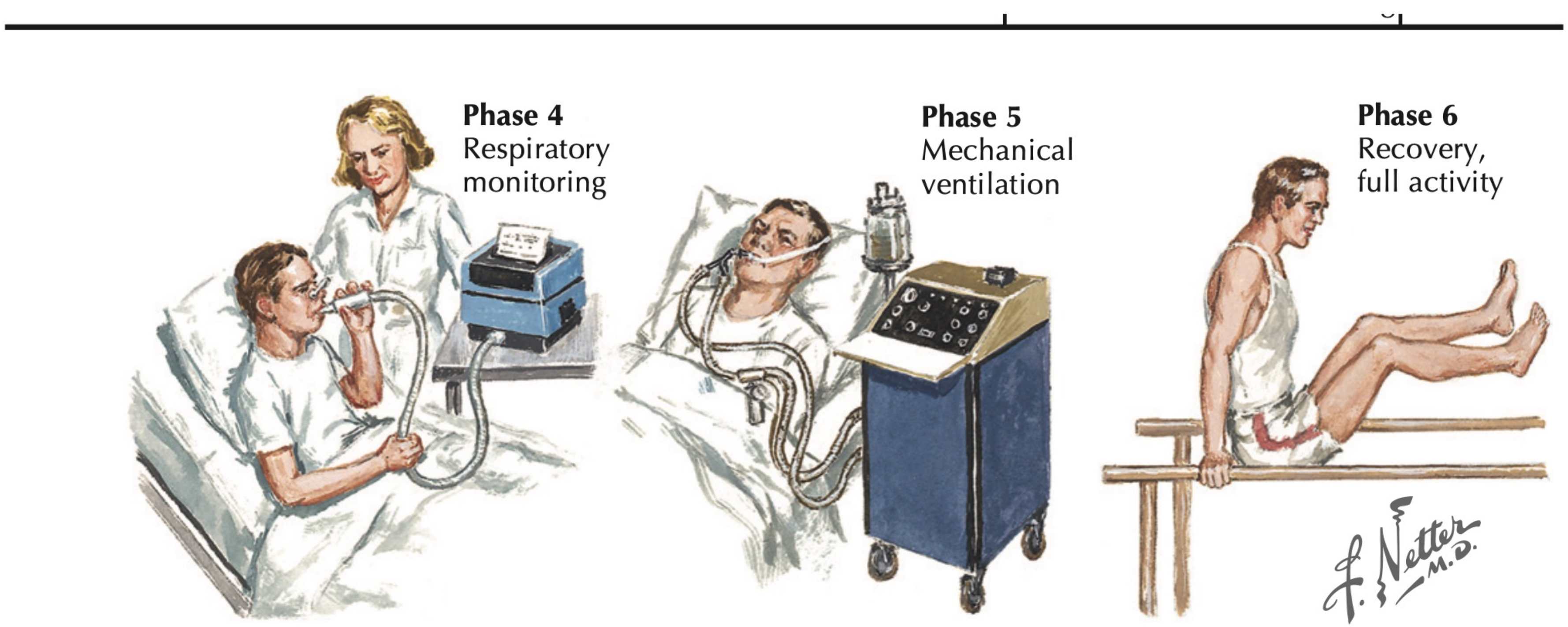


Figure 72-8 Guillain–Barré Syndrome: Electrophysiologic Findings and Clinical Manifestations.

- Corticosteroid therapy has been shown by RCT to be ineffective.
- plasma exchange and intravenous immunoglobulin therapy shorten the duration of ventilation and improve prognosis, provided treatment is started within 14 days of the onset of symptoms
- IVIg and PE are equally effective in reducing the severity and duration of Guillain-Barré syndrome
- No advantage in combining the two treatments.

Prognosis

Overall, 80% of patients recover completely within 3-6 months

4% die

The remainder suffer residual neurological disability which can be severe.

Adverse prognostic features include older age, rapid deterioration to ventilation and evidence of axonal loss on EMG.

Plexus Lesions

- **Brachial plexopathy :** Trauma usually damages either the upper or the lower parts of the brachial plexus, according to the mechanics of the injury. The clinical features depend upon the anatomical site of the damage .
- Lower parts of the brachial plexus ➔ infiltration from breast or apical lung tumours (Pancoast tumour) and may be damaged by therapeutic irradiation.
- ➔ Or anatomical anomalies at the thoracic outlet, which may be accompanied by circulatory changes in the arm due to subclavian artery compression.

- An acute brachial plexopathy of probable inflammatory origin may present with **'neuralgic amyotrophy‘.**

- ➔ period of very severe shoulder pain precedes the appearance of a patchy upper brachial plexus lesion, often affecting the long thoracic nerve which produces winging of the scapula.
- Recovery occurs over months and is usually complete.

- **Lumbosacral plexopathy** ➔ neoplastic infiltration or compression by retroperitoneal haematomas in patients with a coagulopathy.
- A small vessel vasculopathy can produce a lumbar plexopathy, especially in elderly patients when it may be the presenting feature of type 2 diabetes mellitus **(diabetic amyotrophy)** or a vasculitis.
- This presents with painful wasting of the quadriceps with weakness of knee extension and adduction, and an absent knee reflex.

Spinal Root Lesions (Radiculopathy)

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- Most commonly caused by compression at or near their spinal exit foramina by prolapsed intervertebral discs or degenerative spinal disease .
- ➔ infiltrated by spinal and paraspinal tumour masses and inflammatory or infective processes.
- The clinical features include muscle weakness and wasting and dermatomal sensory loss with reflex changes that reflect the pattern of roots involved.
- Pain in the muscles whose innervating motor roots are involved is usually a prominent feature.

Conclusion

- Peripheral nerve and NMJ diseases are common, and they are encountered on daily basis in clinical practice of clinicians of different specialties.
- We should know different presentations of these diseases.
- General physicians need to know their basic and emergency managements.